Outline

Topic	Sub topics	Module
Introduction	CIA Triad, vulnerabilities	Introduction
Database Vulnerabilities	SQLI, prevention	System
System Vulnerabilities	Buffer overflow, malware	System
System Security Practices	OS Security, Software Security	System
Network Vulnerabilities	Scanning sniffing hijacking	Network
Denial of Service	DoS	Network
Security Devices	Firewalls, IDS/IPS, Honeypot	Network
Cryptography	traditional, cryptanalysis, modern (symmetric, asymetric, stream, block)	Mechanisms
Integrity	hashing, MAC, signature, Blockchain	Mechanisms
Authentication	replay attack, challenge response, zero knowledge	Mechanisms
Network Security Protocols	TLS, IPsec, WPA	Network